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Nanoindentation mapping of mechanical properties of cement paste and natural rocks

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Abstract

This paper reports a study to assess nanoindentation mapping of mechanical properties of cement paste and natural rocks. Initial work seems to show that mechanical property mapping by nanoindentation is feasible and can be related to microscopic information. Further work is however required on the effect of indent size and spacing. Such a testing technique can be very useful for materials with different phases to study the intrinsic properties of each component, and also the interaction and properties of the interfacial regions of different phases. The values of Young's modulus and hardness of the individual mineral phases were also determined by statistically analysing a large number of experimental data.
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1. Introduction

Cement based materials and natural rocks are among the most utilized materials, essential to the construction industry and built environment. Studies by microscopic techniques, such as SEM, TEM and image analysis, etc, have revealed that cementitious materials are complex heterogeneous composite materials, with a random microstructure at different length scales, from the nano to the macroscale. Such complex composites are made even more complicated by the time dependent nature of the cement hydration processes which begin at the mixing of cement clinker minerals with water and continue for months and even years. Similarly, natural rocks are essentially composites made from minerals, containing structural features down to the micro/nanoscale.

The engineering properties and durability performance of these materials at the macroscale are all significantly affected, if not dominated, by their structural features and properties at the micro/nanoscale where the deterioration and failure process starts. Traditionally, however, cement paste and rock materials have been treated more or less as homogeneous materials with uniform mechanical properties since mechanical characterisation of such materials has been possible only at a macroscale. It is not until recent years that progress and improved availability of advanced instruments and characterisation techniques, such as depth-sensing micro/nanoindentation, has made it possible to study mechanical properties of various

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micro/nanoscale features in cement-based materials and natural rocks. A few recently published works utilise nanoindentation, and it is still at the early stages in studying cementitious and rock materials. For example, reported research includes investigation of properties of interfacial transition zone in concrete [1–3], of synthetically prepared pure cement clinker phases [4] and selected hydrate phases (e.g. CSH) in well hydrated cement paste [5,6]. A recently published work has also used nanoindentation to determine mechanical properties of minerals related to the classic Moh's hardness [7].

The high spatial resolution of the nanoindentation test offers a scope for assessing the link between mineralogy/morphology and the mechanical properties in multiphase materials. Indeed, nanoindentation mapping of mechanical properties has been applied to study WC-Co alloy [8] and human teeth [9,10]. However, it appears that no such effort has been published to correlate micromechanical properties of specific areas and their microstructural/mineralogical features in cement and rock materials. This paper reports a preliminary study to assess nanoindentation mapping of micromechanical properties of cement paste and natural rocks, and to correlate mechanical properties of different phases with the mineralogy and morphology obtained from optical and scanning electron microscopy. In addition, properties of high and low density CSH and the individual rock-forming minerals were also studied.

2. Materials and experimental methods

Samples of a hydrated cement paste and quartzite and granodiorite rocks were selected for the study. The cement paste was made with a class of 42N ordinary Portland cement and a water-to-cement ratio of 0.35. The cement samples were cured in a water bath (20±3 °C) for 28 days before removal in preparation for test specimens for depth-sensing nanoindentation. Small specimens of the size of approximately 10×10×20 mm were cut from the large cement paste and rock samples prepared using a diamond saw. This was then followed by procedures [1] including resin embedding (but not impregnating), precision sectioning, grinding, polishing (down to 1 μm) and ultrasonic cleaning to obtain the final disc specimens (ϕ30×15 mm) for the nano/microindentation testing. Methanol-based liquids were used as lubricant in the polishing process and for ultrasonic cleaning so as to avoid further cement hydration or possible dissolution of minerals/hydrates.

The quartzite examined was a sample of the Lower Cambrianage basal quartzite, within the Eriboll Formation, from the Assynt area of North West Scotland [11], sampled from an outcrop adjacent to the A835 road at Braes of Ullapool, held in the geological specimen collections of the University of Paisley. It has an arkosic composition, consisting of quartz (SiO_2 approx.60%), and alkali feldspars: potassium feldspar ($K_{0.91-0.96}Al_{0.97-1}Si_{2.98-3.03}O_8$ approx. volume 25%) and sodium feldspar ($Na_{0.82-1.04}Al_{0.97-0.99}Si_{3.01-3.07}O_8$ approx.10%), in addition to minor amounts of muscovite ($KalSi_5O_{10}(OH)_2$) and pyrite (FeS_2). It is fine grained (typically between 100 and 250 μm) with generally equant crystals with mostly planar sutured contacts in a subpolygonal recrystallised texture. There is a slight preferred orientation of crystals, which can be seen on SEM-BSE images, picked out by the elongation of potassium feldspars.

The granodiorite is from the Criffel Igneous complex of South West Scotland, a large intrusive igneous body of Devonian age. The sample studied is a course grained rock (≤5 mm) dominated by Potassium Alkali and Sodium Plagioclase feldspars, (these latter, in the area indented are Oligoclase $Ca_{0.16}Na_{0.80-0.83}K_{0.01}Al_{1.18-1.20}Si_{2.77-2.8}$ $N=8$, determined from quantitative EDS analysis), in addition to strained polycrystalline quartz, biotite, apatite ($Ca_5(PO_4)_3(OH, F, Cl)$) and some carbonate minerals. Many of the feldspars appear intergrown with Fe–Mg rich muscovites (5.5% FeO+MgO). The feldspars also exhibit twinning and compositional zoning, though this was not sampled in the present work. Characterisation of the rock type was limited for the present study to the minerals indented as part of our experiment (see below). Our intention in this study was to evaluate the nanoindentation-based micromechanical property mapping; not to explicitly characterise the samples for the purposes of geological or other interpretive purposes.

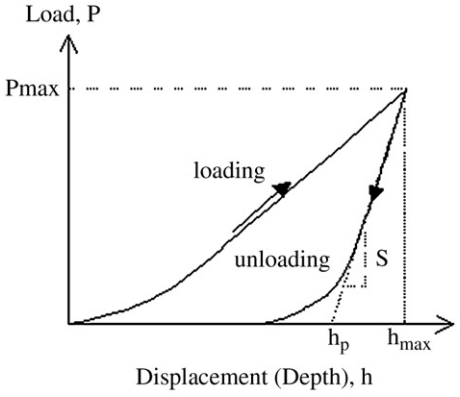


Fig. 1. A schematic representation of a typical indentation load-depth hysteresis curve.

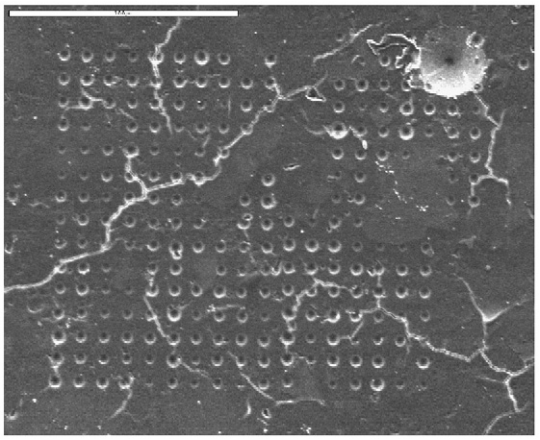


Fig. 2. Secondary electron image of the area of indents on the cement sample studied. Areas of harder phases, such as clinker grains are easily identified. Note how the indent size, or even the presence of a *visible* indent, varies with phase type, in response to mechanical property.

The methodology and operating principle for the nanoindentation technique have been reviewed and presented in detail elsewhere [12,13]. A typical outcome of the nanoindentation testing is an indentation load-depth hysteresis curve as shown in Fig. 1. As a load is applied to an indenter in contact with a specimen surface, an indent/impression is produced which consists of permanent/plastic deformation and temporary/elastic deformation. Recovery of the elastic deformation occurs when unloading is started. Determination of the elastic recovery by analysing the initial part of the unloading data according to a model for the elastic contact problem leads to a solution for calculation of the elastic modulus E and hardness H of the test area.

The nanoindentation apparatus used in this study was Nanoindenter XP(MTS Systems Corporation). In this study, all testing was programmed in such a way that the loading started when the indenter came into contact with the test surface and the load maintained for 30 seconds at the pre-specified maximum value before unloading. Progressive multistep indentation testing [14] with five loading–unloading cycles at different depths was applied at each test point. The unloading data for the lower indentation depth (i.e. h_p=300–400 nm) was used to determine the modulus and hardness values of the indentation point, whereas the indent impression produced at the higher depth (h_p=1000–1500 nm) allows easy examination of the indent and the tested area after the testing using optical and scanning electron microscopes (SEM).

Two different shapes of diamond indenter were selected for this study. A common Berkovich indenter, which has a relationship between the projected area A and the depth h_p for the indentation as $A \approx 24.5h^2$, was used for studying the rock samples. A conical indenter with $A \approx 4.44h^2$ was used for studying the cement paste sample as it copes better with the rougher surface and smaller-scale spatial details often seen in hydrated cement specimens. The comparatively sharper conical indenter allows smaller spacing between indents (so to

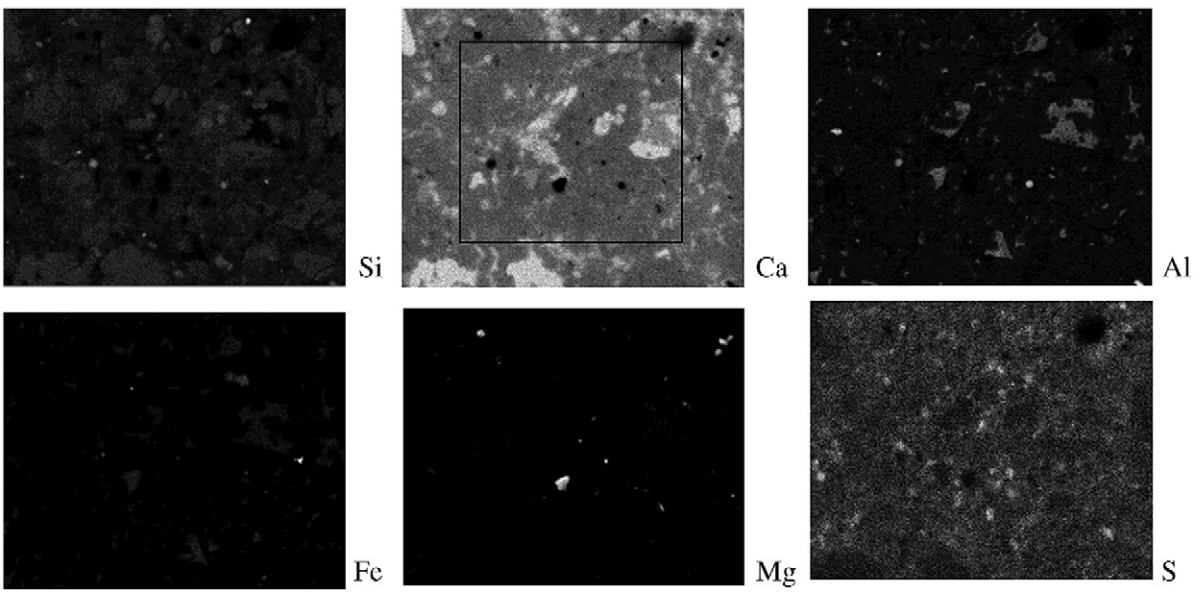


Fig. 3. Element distribution maps o f the area of indents on the cement sample, permitting the clearer identification of phase constituents in the cement paste, such as unhydrated clinker, calcium hydroxide, C–S–H. The rectangle in the Ca map indicates approximate area of indents.

avoid possible overlapping of indented areas) to be used as well. Information on the mechanical properties was obtained from a matrix of a minimum of 240 indents covering a representative area of at least $150 \times 150\ \mu m^2$ and $300 \times 300\ \mu m^2$ on the surface for cement and rock specimens, respectively. The selected indent spacing was 10 μm and 20 μm for the cement and rock specimens, respectively. The tested areas in each sample, including the matrix of indentation, were then examined using an optical microscope utilising reflected light. The samples were also studied using a FESEM (Hitachi S4100), in both secondary electron (working distance of >15 mm, accelerating voltage between 5 and 20 kV) and backscattered electron (BSE, 20 kV) imaging modes, in conjunction with energy-dispersive spectroscopy (EDS) analysis for the determination of qualitative and quantitative mineral compositions (cobalt gain and natural mineral calibration was applied). These enable the assessment of correlation between the nanoindentation results and the visual appearance of the corresponding surface (i.e. morphology), as well as the mineralogy and composition of the phases involved.

3. Results and discussion

3.1. Mapping of mechanical properties

Fig. 2 shows a secondary electron image of the area of indents on the cement sample. Fig. 3 shows the element maps, determined by EDS, for a portion of the corresponding area (within the rectangle on the Ca map), that allows the identification of phases, and Fig. 4 presents the corresponding mechanical property maps. It appears in these figures that the Young's modulus map and the hardness map have a similar appearance. The unhydrated, or partially hydrated, cement clinker particles can be identified as a dark grey shade in the optical image, whereas the hydrate phases are a lighter grey shade, confirmed by the EDS elemental mapping presented in Fig. 3. The EDS mapping also permits the identification of areas of calcium hydroxide, and offers a higher resolution, permitting an identification of the phases indented. Viewing the modulus/hardness map and the optical and SEM images together, the *general* correlation between the mechanical properties and phase features can be clearly seen. For example, it is apparent that the highest value of modulus/hardness is in the core of the large unhydrated/partially-hydrated cement particles, and the property values rapidly decrease outwards and into the space filled by the cement hydrate phases which are reported to have modulus values in the range of 20–40 GPa [5,6]. The lowest values of modulus (i.e., 2–20 GPa) are believed to be due to the porosity or surface defects present in the space originally occupied by mixing water.

Similarly, for the quartzite and granodiorite samples, maps of elastic modulus and hardness of the tested area have a similar appearance (Figs. 5 and 8). These properties maps can also be seen to correlate well, on a qualitative level, with the mineralogical features of the optical and SEM images of the corresponding areas. Fig. 6

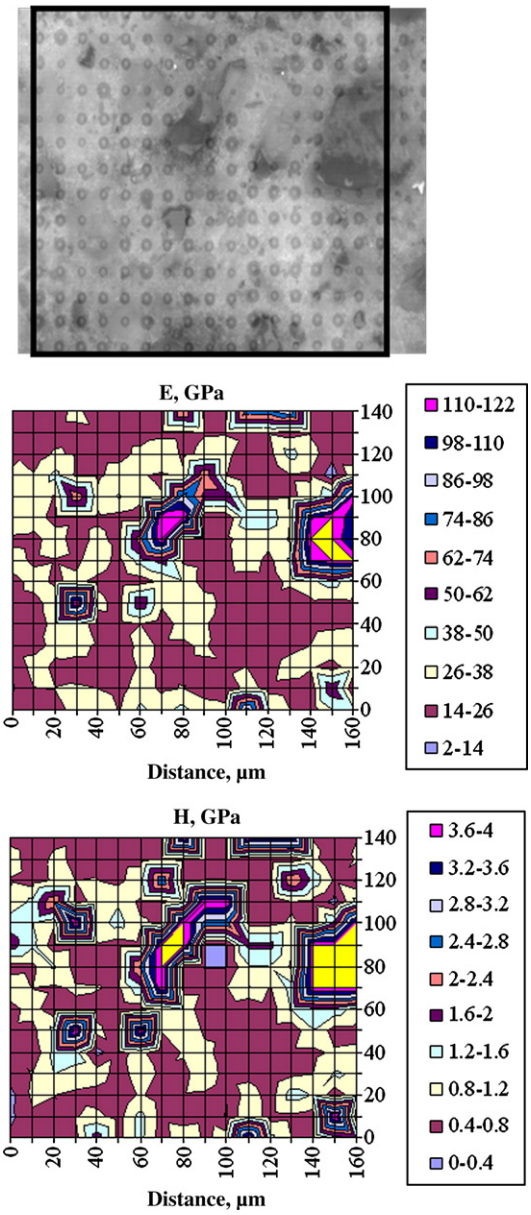


Fig. 4. Maps of mechanical properties (elastic modulus *E* and hardness *H*) of the test area in cement paste, together with the corresponding optical microscopic image.

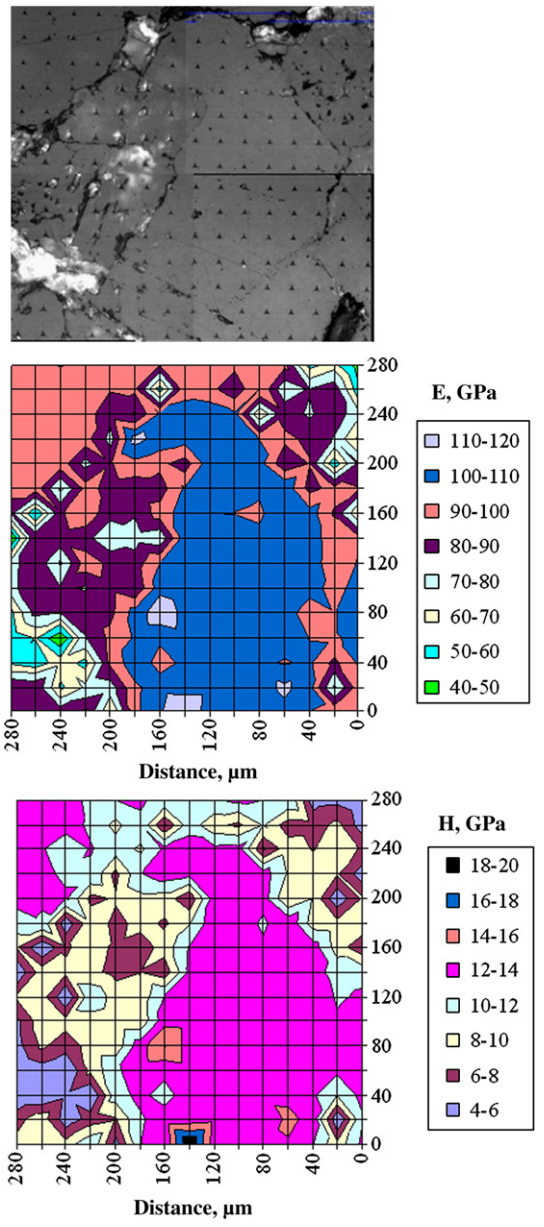


Fig. 5. Maps of mechanical properties (elastic modulus *E* and hardness *H*) of the test area in quartzite, together with the corresponding optical microscopic image.

presents an element distribution map for the tested area in the quartzite sample, which permits the identification of quartz (the area with no Al, K or Na) and also K and Na feldspars, orthoclase and albite. The area of quartz corresponds closely to the area with consistent modulus values of 100–110 GPa and hardness of 12–14 GPa. The orthoclase feldspar corresponds to areas with modulus values of 80–90 GP and hardness values of 8–10 GPa.

Fig. 7 is a BSE image of the area of indents on the granodiorite, which reveals a high apparent heterogeneity with in the feldspar grains. The image confirms the presence of Na rich Plagioclase feldspars, revealed to be Oligoclase from quantitative EDS analysis in the SEM), with intergrowths of a fine grained Mg/Fe-containing muscovite mica, areas of calcite and also Apatite. Element mapping was also carried out, but is not presented here as the BSE image summarises the composition of the sample adequately. The BSE image also reveals useful detail of the locations of indents, in this case as there is apparently a significant SE contribution to the sample image.

For the granodiorite sample, the general comparison of the mechanical property maps (Fig. 8) and the mineral phase distribution is clear , but there appears to a considerable variation in property across the majority of the sample. This is primarily due to areas of an intergrowth of feldspar with relatively fine-grained muscovite within the sample (Fig. 7). The area of low modulus to the upper left of the sampled area corresponds to an area of Mg–Fe containing muscovite, a sheet silicate mineral with low values of mechanical property. High modulus areas to the right of the area sampled correspond with grains of Apatite.

It is also apparent that the variation of mechanical properties was smaller in the quartzite and granodiorite samples than in the cement paste sample. This is due to the detailed nature of the *distribution* of phases, and *defects* in the sample. For the granodiorite sample, though it is a coarse-grained rock, the nature of the area sampled, with many small grains of mica within the feldspar grains (areas labelled “Mix” in Fig. 7), means that the mapped values of modulus and hardness reflect a heterogeneity that is of a similar scale to the density of sampling, therefore producing a material property map that is undulating. The method of contouring the values of mechanical property also tends to emphasise a transition in values that does not reflect the discrete spatial nature of mineral grains. For the granodiorite sample, the size and spacing of indents should be decreased and number of indents increased to improve the resolution of the mechanical property maps. The phase grain size in the quartzite specimen is significantly larger than those in the cement paste sample, and the effective size in the granodiorite sampled, therefore producing a “flatter” property map than either of these two.

3.2. Determination of mechanical properties of individual hydrate and mineral phases

With the large number of indentation test results generated for the mapping of mechanical properties,

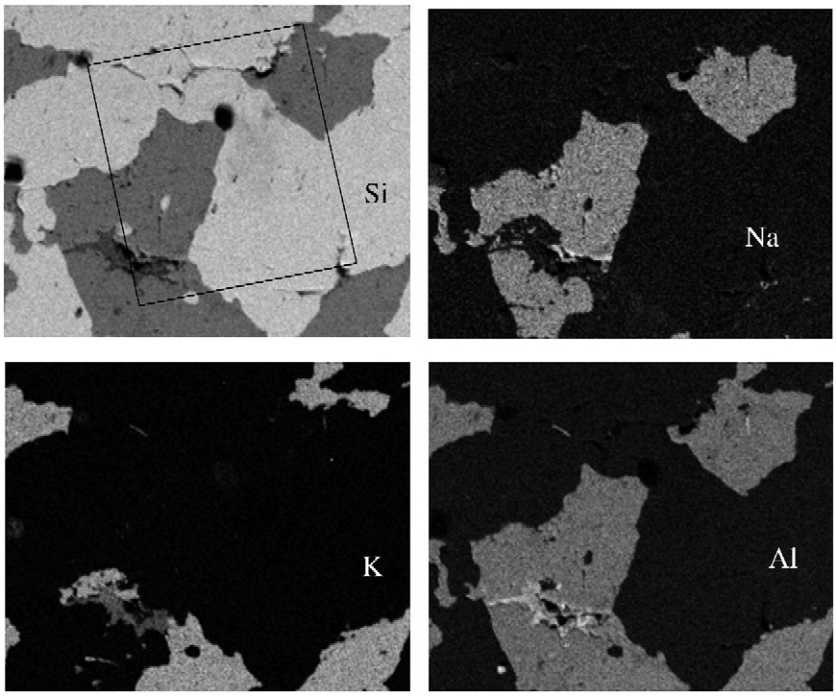


Fig. 6. Element distribution maps of the area of indent in the quartzite sample, highlighting the presence of quartz, Na and K feldspars in the area of sampling. Note also the presence of surface defects in the sample (cracks and voids) that can affect the quality of data obtained. The inclined rectangle in the Silicon map indicates the approximate area of the indents.

attempts were also made to extract mechanical properties of specific individual phases in the tested area by statistically analysing all the test results, using a method similar to that presented by Constantinides et al. [5]. Basically, the experimental data (i.e. modulus and hardness values) were statistically analysed to produce a frequency histogram. Then, the best model fit to the experimental results with multimodal normal distribution curves (also known as Gaussian distribution, see Eq. (1)) was produced using nonlinear least squares method. These curves are shown in Figs. 9–11.

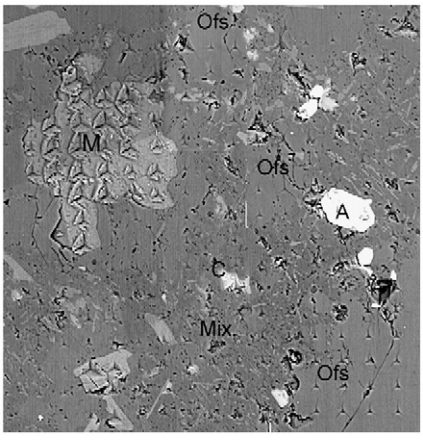


Fig. 7. BSE image of the area of indent in the granodiorite sample. This should be viewed along with Fig. 8. The image is dominated by an oligoclase plagioclase feldspar (Ofs), which constitutes the entire background to the sample area. The sample also contains a Mg/Fe (~5%) containing muscovite (M), and minor Calcite (C) and Apatite (A). The texture is mixed in many areas (e.g. Mix) with intergrowths of fine grained muscovite and feldspar, and many of the indents sample more than one mineral. In the lower right and top centre areas there is feldspar with no muscovite intergrowth.

$$f(x;\mu,\sigma) = \frac{1}{\sigma\sqrt{2\pi}}\exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) \quad (1)$$

From each model fit, the mean value μ and standard deviation σ of the distribution were extracted, whose association with a specific hydrate or mineral phases was ascertained by SEM-BSE and SEM-EDS analysis [6]. The area under the normal distribution curve could also provide an estimate of the volume fraction for the hydrate/mineral phase it associated with, within the area of the sample covered by indents.

Results for the cement paste in Fig. 9 were obtained by using only experimental data considered to be associated with cement hydration products (i.e., in the range of 10–40 GPa and 0.1–1.7 GPa for modulus and hardness, respectively). It was not difficult to separate this data group from the entire data set as the experimental

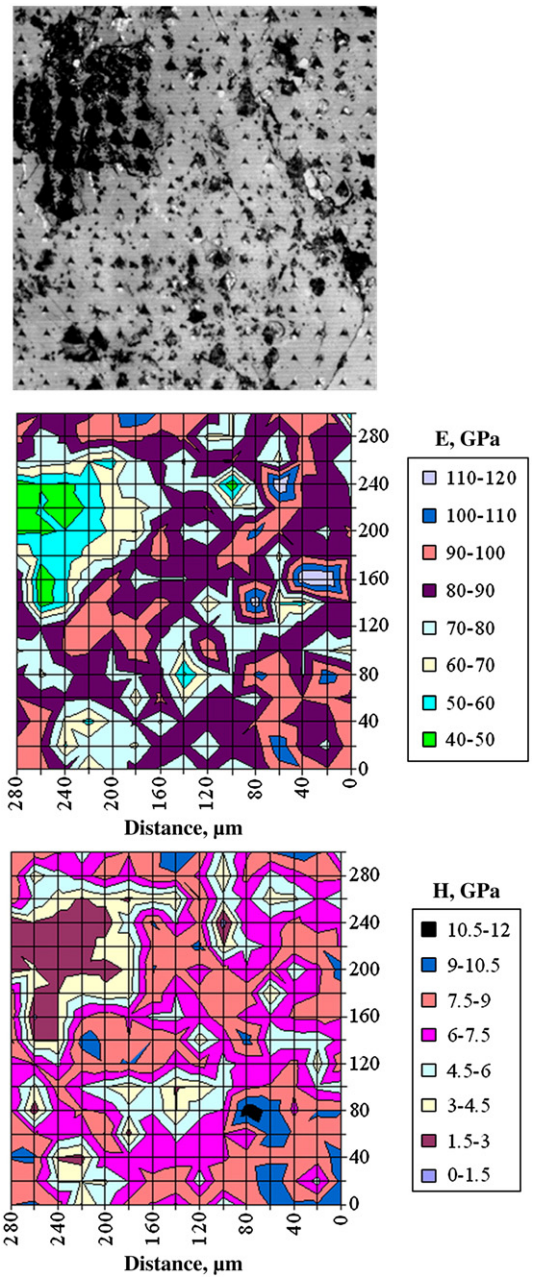
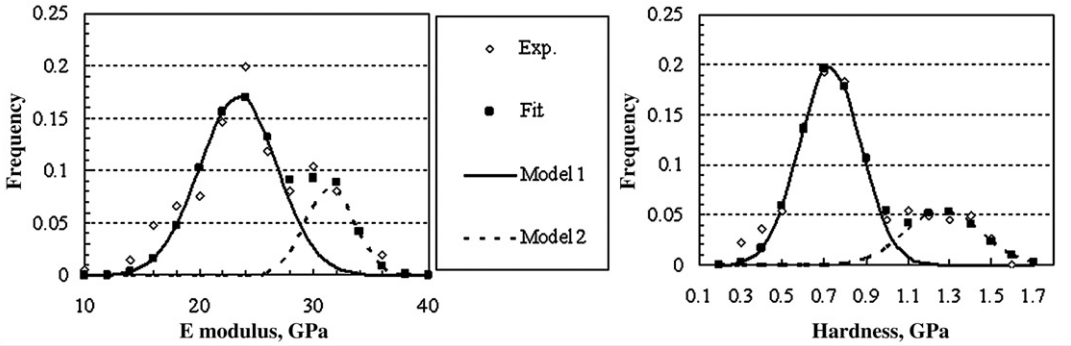


Fig. 8. Maps of mechanical properties (elastic modulus E and hardness H) of the test area in granodiorite, together with the corresponding optical microscopic image.

E and H values associated with the unhydrated/partially-hydrated clinker phases were much higher and no significant overlapping for the two data groups was present. As shown in Fig. 9, two modal normal distribution curves could be used to produce a good fit to the experimental data. Analysis by SEM-BSE and EDS along with comparison with the modulus and hardness maps, allowed a further corroboration to be made between the two main hydration products; low density and high density calcium silicate hydrates (i.e. LD-CSH and HD-CSH), and the modelled curves. The mechanical property values extracted for the two phases are in very good agreement with the values reported by Constantinides et al. [5] (e.g. $E_{\text{LD-CSH}}=21.7\pm2.2$ GPa, and $E_{\text{HD-CSH}}=29.4\pm2.4$ GPa) and by Acker [15] (e.g. $E_{\text{LD-CSH}}=20\pm2$ GPa, and $E_{\text{HD-CSH}}=31\pm4$ GPa). This further supports the argument that the elastic properties of the two types of CSH are intrinsic to cement paste [5].

Fig. 10 shows the experimental data and the model fitting curves as well as mechanical properties extracted for the individual mineral phases identified for the quartzite specimen. Clearly, three model normal distribution curves are required to produce the best fit to the experimental data. SEM-BSE and EDX analysis further identified three mineral phases present in the tested area as Albite, Orthoclase and Quartz. Thus, the E and H values extracted from the fitting curves could be linked with the individual mineral phases present, as shown in Fig. 10. Broz et al. [7] has reported E and H values of orthoclase and quartz obtained by carrying out micro/nanoindentation testing of specially prepared pure phases of such minerals. For the orthoclase mineral their reported E and H values are 89.2 ± 7.1 GPa and 9.11 ± 0.58 GPa, respectively. For the quartz mineral, their reported E and H values are 117.6 ± 2.7 GPa and 14.54 ± 0.42 GPa, respectively. It is interesting to note that the E and H values for orthoclase in the quartzite sample (as determined in this study) are in excellent agreement with those reported. For quartz, the agreement of our values and the reported values is reasonable, but not as good as in the case for orthoclase. It is believed that different crystal orientations of the an isotropic minerals tested might have contributed such differences in properties.

Similar to the quartzite sample, three model normal distribution curves were also used to fit the experimental data for the granodiorite sample. The three mineral phases identified in the test area and their E and H values extracted from model fits are shown in Fig. 11. There is clearly no quartz present in the area of the granodiorite sampled. There is a mineral with low E and H values (59.2 ± 8.1 and 3.3 ± 0.6 GPa, respectively) which is identified by BSE and EDS analysis as a Mg/Fe-containing muscovite. The third model curve is fitted to values of E and H of 91.6 ± 6.5 and 8.6 ± 1.1, and this can be correlated through the images and mechanical property maps to be oligoclase feldspar. The middle values of modulus fit a curve with values of E and H of 77.6 ± 5.1 and 5.8 ± 0.9. Comparing the



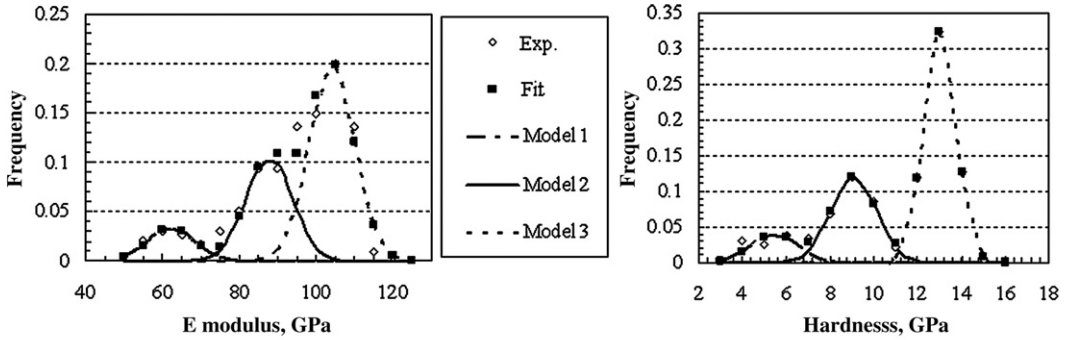
Properties of cement hydrates identified		Modulus, GPa	Hardness, GPa	Volume, %
Model 1	LD - CSH	23.4 ± 3.4	0.73 ± 0.15	75.9
Model 2	HD - CSH	31.4 ± 2.1	1.27 ± 0.18	24.1

Fig. 9. Mechanical property frequency histograms for test results of cement hydrates, together with the *E* and *H* values extracted from model fits for the individual hydrate phases identified.

mechanical property maps and the BSE images of the granodiorite sample, it is apparent that this middle value curve represents values from indents formed feldspar crystals with a significant intergrowth of muscovite. Many indents can be seen to impact at the interface between these two minerals. It is possible also to argue that the property values presented for the intermediate value curve appear to sit midway between the pure mineral phases, though it would not be possible to confirm the quantitative nature of the mixing effect without further analysis.

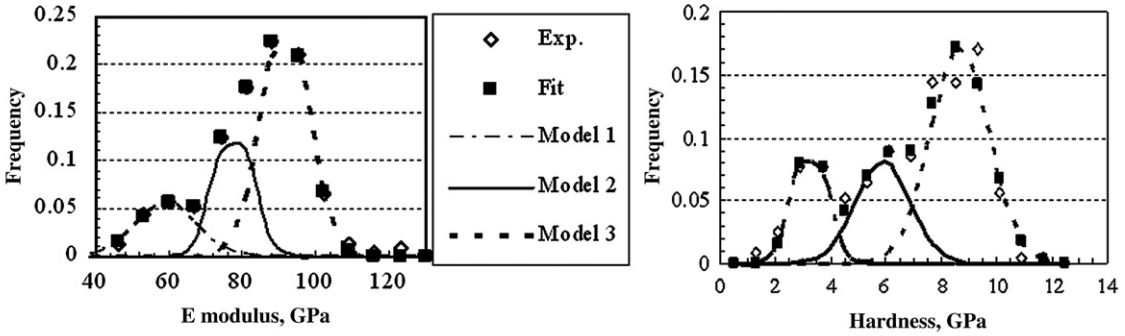
3.3. General discussion

With the use of the progressive multistep indentation tests (five cycles in this study), five sets of modulus and hardness values were obtained at each test point (each one at a different depth). This made it possible to assess the effect of indentation depth or length scale (i.e., size effect) on measured properties. Determination of mechanical properties at reduced indentation depth would potentially allow the use of smaller indent spacing and higher spatial resolution in test/mapping. The results



Properties of mineral phases identified		Modulus, GPa	Hardness, GPa	Volume, %
Model 1	Albite ($NaAlSi_3O_8$)	62.2 ± 6.0	5.4 ± 1.0	10.0
Model 2	Orthoclase ($KAlSi_3O_8$)	87.8 ± 6.0	9.1 ± 1.0	31.6
Model 3	Quartz (SiO_2)	104.2 ± 5.9	13.0 ± 0.7	58.4

Fig. 10. Mechanical property frequency histograms for test results of quartzite, together with the *E* and *H* values extracted from model fits for the individual mineral phases identified.



Properties of mineral phases identified		Modulus, GPa	Hardness, GPa	Volume, %
Model 1	Muscovite	59.2 ± 8.1	3.3 ± 0.6	16.7
Model 2	Mixed oligoclase/muscovite	77.6 ± 5.1	5.8 ± 0.9	25.8
Model 3	Oligoclase feldspar	91.6 ± 6.5	8.6 ± 1.1	57.5

Fig. 11. Mechanical property frequency histo grams for test results of granodiorite, together with the *E* and *H* values extracted from model fits for the individual mineral phases identified.

obtained from this study, however, appeared to show that at the smaller indentation depths of ~100 nm both *E* and *H* values were significantly affected by the surface roughness of the specimen and showed much increased variations, particularly for the cement sample. Therefore, the specimen preparation for such materials may have to be improved if indentation depths smaller than 100 nm are required. Currently, with the available instrument and the use of the steep conical indenter, the minimum indent spacing possible is probably 3–4 μm. For materials with relatively large grain size, such as quartzite and granodiorite used in this study, the results appeared to show very small variations in properties within the same mineral grain. Thus, an increase of indent spacing for such materials may be more beneficial as for the same number of indents it would allow a larger, more representative surface area to be studied/mapped. Further work is clearly required in this aspect.

The creation of maps of mechanical properties using nanoindentation can be useful in many applications. As in this study, it can be used on materials with different phases to study the intrinsic properties of each component. It is also valuable to study the interaction and properties of the bond and interfacial regions of different phases. Furthermore, mechanical property maps and their improved link with microstructural features can significantly support realistic image-based modelling.

However, the mapping method interpolates contours and implies intermediate values in areas away from the points of testing. It has been shown that minerals, especially in some natural rocks with a relatively large grain size, will have consistent values within grain boundaries, though not in all cases. This means that it is likely that the interpolated, contoured values shown on the mechanical property maps are artefacts of interpretation and do not necessarily (and probably do not) represent the real distribution of properties at a higher sampling resolution. Minerals are mostly discrete granular entities within materials. Correct modelling of properties that utilises indentation mapping requires a higher resolution of spatial information to remain accurate. This emphasises the need for detailed, thorough microscopy to accompany and corroborate the indentation mapping. With further work (perhaps including targeted indentations on minerals, in addition to grid-like sampling) it should be possible to establish a method to map properties to a high resolution, and up to the interfaces between grains. The interfaces between mineral grains within composite materials are central to an understanding of the physical and mechanical behaviour of bulk materials, and should properly be an area of study using the indentation method.

Values for intrinsic mechanical properties, such as hardness and elastic modulus, of individual phases in cementitious and geo-materials have particular relevance in construction, geology and materials research. For example, weathering and degradation of cementitious and geo-materials have significant impact on durability of buildings/infrastructures and conservation of historical monuments. However, the relationships between chemical, biological and physical weathering/degradation of these materials and the mechanical properties of

the specific constituent phases responsible are poorly understood. The high spatial resolution of the nanoindentation method appears to be ideally suited to the study of the fundamental behavioural factors related to material durability and deformation (e.g., creep and shrinkage) at the macroscale.

4. Conclusions

1. The results of this study showed that the mapping of mechanical properties of cement paste and natural quartzite and granodiorite samples by nanoindentation is feasible and can be related to microscopic information. Results from such tests can also be used to study the intrinsic properties of each component, as well as the interaction and properties of the interfacial regions of different phases for materials with different phases. Potentially, the mechanical property maps can be very useful in supporting realistic image-based modelling.
2. Further work is, however, required to study the effect of the indent size and spacing so that optimum selection of these parameters could be made for the specific material in order to study a representative materials area and volume, and to provide mechanical property maps with high spatial resolution.
3. The values of Young's modulus and hardness of the individual hydrate and mineral phases determined by the statistical method appeared to be in good agreement with the limited published data.

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